

Energy Efficiency Improvements in Italian Historical Buildings:

The case study of Ascoli Piceno

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Abstract— Evaluating the energy efficiency of historical buildings is an important theme nowadays, with European directives that fix a strong reduction in energy use and impose an increase in the amount of energy produced by renewable sources. Within restoration and other related disciplines, it is therefore necessary that a specific method of investigation for historical buildings be developed. The methodological approach of energy efficiency in historical buildings, from the building scale to the urban scale, must account for the particular characteristics of historical buildings. Each operation should be carefully examined in each case and defined within a grid of possible interventions that can combine conservation and innovation. Public administrations have also recently addressed these themes to evaluate medium and long-term strategies useful in more effectively managing the heritage.

These objectives form the basis for recent research involving the city of Ascoli Piceno. The aim is to provide the entities that manage the cultural heritage with criteria and methods to completely and critically assess historical buildings while also duly considering aspects related to their energy performance.

Keywords— Cultural heritage; Energy saving; Sustainability

I. INTRODUCTION

The increase in international protocols and the recent publication of Italian guidelines to improve energy efficiency in the cultural heritage show the importance of energy efficiency. This is especially true when defining effective interfaces between technological tools and conservation principles in order to identify a way to rationalize resources. These interfaces therefore serve as tools (mathematical, diagnostic, and meta-planning) that allow methodological indications to be defined.

Correct energy diagnosis represents a fundamental step in renovating buildings. A modification of the architectural organism could sometimes be required, which, if not carefully planned, would lead to problems that could then negatively affect the monumental and/or documentary value of the building [1]. The development of such a diagnostic procedure is certainly not simple, since it requires an analysis capacity that should be developed and applied within the specific field of restoration. Currently, investigations regarding aspects related to forms of energy production address the application of

innovative technologies only in contexts outside of historical buildings [2]. For this reason, this research focuses on a method oriented at defining retrofitting strategies that can be applied to the historical city and, in particular, to a building of notable cultural worth, seeking interventions that can protect its characteristics.

This paper presents the results of an experiment involving the city of Ascoli Piceno. The aim is to provide entities that manage the cultural heritage with criteria and methods to completely and critically evaluate historical buildings while also considering aspects related to their energy performance. All of this also relates to the now unavoidable need for public administrations to rationally contain management costs.

II. ASCOLI PICENO: TOPOGRAPHY AND URBAN DEVELOPMENT OF THE SITE

Ascoli Piceno is a city of 50,000 inhabitants situated in the Marche Region in Italy. It lies in a sub-Apennine region characterized by cold, humid winters with possible snow showers and quick changes in temperature. Summers are hot, with infrequent precipitation. The average temperature in January is 5–6°C, while in July it is 24–25°C.

The first settlement dates to the pre-Roman era and was built on the floodplain created by the two rivers that surround the historical centre: the Tronto and the Castellano (Fig. 1). The city is very uniform with respect to typical materials and forms, with immediately recognizable recurring elements such as towers, campaniles, rooftop terraces, and tiled roofs. Each architectural element corresponds to a communication process conducted within a precise search for an urban image.



Fig. 1 Topography of Ascoli Piceno. The city is very uniform with respect to typical materials and forms, with immediately recognizable recurring elements such as towers, campaniles, rooftop terraces, and tiled roofs.

Within the historical center, one aspect about the public buildings should be highlighted: there are no new constructions except in rare cases, and most public activities are carried out in historical/monumental complexes. Over time, these have been adapted for new uses through numerous non-systematic interventions and therefore many of these buildings still present energy, management, and safety problems.

Ascoli Piceno offers the chance to define a methodology applied to a meaningful historical context. Energy performance, the formalization of a correct framework of requirements, and the effective possibility of savings have been widely debated in the City for some time.

After analyzing the cultural and regulatory approach to energy renovation in existing buildings, the research addressed conservation problems related to the energy efficiency. The methodological setup of the project considered the inseparable conservation/new energy technologies binomial, refining it through scenario analysis to guarantee the indispensable energy performance for the historical heritage according to current regulations, while respecting the historical/architectural values of the city [3]. The approach proposed is exemplified by a specific urban building. It leads to the critical identification of individual factors, but can also be applied to other similar contexts. At present, the view shared by scholars in various disciplines is to consider the possibility that the limited historical/architectural heritage can also be subject to investigations aimed at reaching efficiency objectives, while still assuming the necessary methodological tools for active conservation of the building. To this end, an energy methodology is applied, which can be useful in implementing a series of analyses to reach the predetermined objectives [4].

III. APPLICATION OF THE ENERGY METHODOLOGY

The Palazzo dei Capitani del Popolo in Ascoli Piceno is used as a case study. To analyse the building, different levels of investigation were made corresponding to its characteristics, its construction history, the peculiarities of the materials used, its current and future use, the building's electrical/mechanical systems, and measured consumption [5]. The goal is to obtain an effective energy diagnosis in order to orient choices towards

innovative production systems that allow the energy efficiency objectives to be reached, even for those buildings in which the most important element is the historical/architectural value that has accumulated over time [6].

The first phase of the research was to analyze the historical/architectural characteristics of the building. The ancient basis of the Palazzo dei Capitani del Popolo (Fig. 2) was composed of a house/tower (ancient merchant building from the thirteenth century) and a house with a tower (feudal building from the twelfth century) located in the northwest corner of what was then the small square of the city market.

The square was subsequently transformed and expanded into the current Piazza del Popolo. As with all large monumental complexes, the Palazzo dei Capitani was continually subject to interventions and improvements. In 1296, to allow for its expansion, the above-mentioned buildings were combined into one, aligning them with the addition of a new façade that unified the front on the square. From the middle of the fourteenth to the fifteenth centuries, the palace saw further transformations and the northern part was added. This addition transformed the building into a typical form of medieval palatial structure. To realize the new construction, it was necessary to demolish some minor buildings located on the northern side. These interventions were strongly desired by Popes Julius II and Leo X, who donated 1100 ducats for the restoration works.

Around 1445, the City Elders had to change residence and moved from the Palazzo dell'Arenco to the Palazzo dei Capitani. The modifications were so profound that the original medieval physiognomy of the building was lost. Other important modifications included the addition of the portion expanding from the tower towards the south, and the division of the expansive rooms on the first floor into two separate levels. Between 1518 and 1520, the city magistrates initiated works to remake the back façade of the Palazzo, accepting a project by Nicola Filotesio (known as Cola dell'Amatrice), which was carried out by the Lombard masters, who were particularly active in the city from the beginning of the sixteenth century. From the seventeenth to the nineteenth centuries, Ascoli was characterized by a period of widespread unease and by a general political, socioeconomic, demographic, and cultural decadence. In the first years of the nineteenth century, the Palazzo dei Capitani housed the seat of the vice prefecture, which accelerated the palace's degradation since only some of the rooms were occupied. After the French period, Ascoli again fell under the pontifical government and the new apostolic delegate took up residence in the palace, which had become completely dilapidated. In 1821, the apostolic delegate drew up a project, with estimates for the integral restoration of the palace. In the period 1925–1939, a series of interventions was made with the involvement of the architect Vincenzo Pilotti. As a consequence of pressure from some private citizens and the presumption of the dominant party, three events occurred that were incisive not only for the use of the palace, but especially for its architecture, which entailed serious alterations to the original structures. The City approved Pilotti's project provided that no modifications were made to the exterior or, in particular, to the façade. Immediately after the Second World War, the palace began to



Fig. 2 Piazza del Popolo and the palace before the restoration.

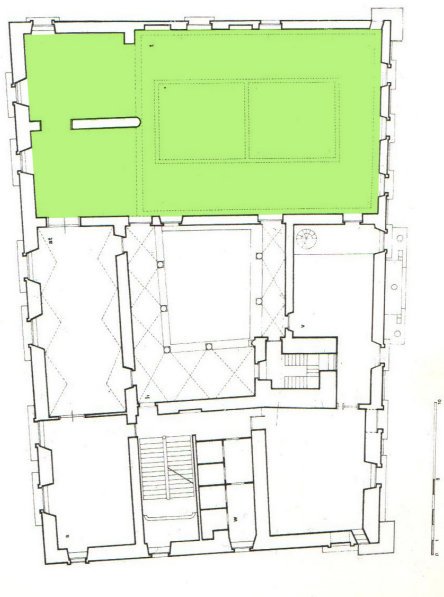


Fig. 3 Sala della Ragione.

show further signs of degradation that also affected its structure. By 1980, even if largely consolidated, the palace was still completely unusable. The administration decided to organize a project to definitively reopen it, entrusting city technicians with the task. The Palazzo dei Capitani was reopened to the public on 10 October 1987, after being closed for 20 years (Fig. 2).

The second phase of the research dealt with analysis of the construction and material characteristics. The building is composed of four storeys plus a basement, which houses the archaeological museum. In the third phase of the research, all the elements of the historical building were analysed (i. e. vertical curtain walls, doors and windows, floors and roofs) [7].

The functions identified within the historical building are particularly varied and are distributed over the various levels. This case study presents a complex building, with offices for cultural services on the first floor and a series of rooms for temporary exhibitions on the second. The third floor houses the Sala del Consiglio Comunale, also known as the Sala della Ragione (fig. 3).

The façade overlooking Piazza del Popolo conserves all the historical stratifications that have occurred beginning with its use as the City Hall in the fifteenth century until its last restoration in 1987.

In the third phase of the research, the following principal elements were analysed [7]:

- Particular elements of the historical building. These elements elude rigid classification in that they determine the character of the building and its uniqueness. Within the compact rectangular body, there is an ample square courtyard that stretches the height of the building, with arcades of rounded arches on three levels that create a particular internal microclimate. This configuration is undoubtedly one of the weak points in the complex and was therefore subject to a series of specific analyses.

- Vertical curtain walls. These were analysed to determine the transmittance, stratigraphy, and thermal capacity of the walls, which are values of fundamental importance. In particular, the building in question has undergone numerous transformations over time. The external envelope is massive and of variable thickness and it contains openings of various sizes in addition to two deep loggias.

- Doors and windows play an important role in the thermal exchange of a building. Construction information was collected regarding the windows, identifying their geometrical dimensions, materials used, closure systems, and other indications useful in defining their state of conservation and efficiency.

- Floors and roofs were in part substituted with reinforced concrete slabs, in accordance with the regulations in effect when the restoration was made. There remain, however, a series of vaulted ceilings in brick of limited height, especially in the lowest levels, while in the Sala della Ragione a large, fully-equipped false ceiling was installed.

IV. DISCUSSION

The scope of these investigations is to understand the current energy state of the building to orient design strategies to improve performance.

For the case study, monitoring the building and its systems consisted in collecting, processing, and interpreting quantitative and qualitative information about its characteristics and operation [8]. Thermophysical information was collected using non-destructive techniques, such as infrared thermography and thermal mass flow. For a historical building, optimal initial results can be obtained by using thermography for historical analysis, structural analysis, study of the state of conservation of the framework, and energy diagnosis [9]. Thermal mass flow investigations were then made to assess the energy efficiency of the building.

Thermophysical information was collected using non-destructive techniques, such as infrared thermography and thermal mass flow. Thermography allows the apparent surface temperature of bodies to be mapped, measuring the presence of anomalies in the surface thermal distribution. For a historical building, optimal initial results can be obtained by using thermography for historical analysis, structural analysis, study of the state of conservation of the framework, and energy diagnosis (Fig. 4) [9].

To characterize this specific building/system as a whole, thermographic analysis of the structure was made to identify thermal bridges in the exterior envelope and to verify the heating system within.

Thermographic readings did show, however, that thermal dispersion is found near windows and doors, which is due to both the presence of traditional glass and imperfect closures, which, while not being historical, are obsolete with respect to current norms. In addition, the doors and windows do not have shading systems either inside or outside in respect for the monumental character of the building.

Thermal mass flow investigations were then made to assess the energy efficiency of the building (Fig. 5).



Fig. 4 Thermographic analysis of the Palazzo dei Capitani del Popolo east facade.

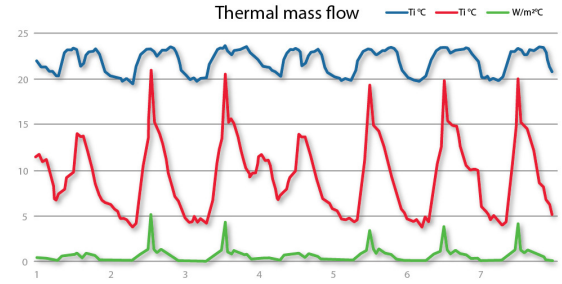


Fig. 5 Thermal mass flow investigations

The investigations consisted in installing probes (flow sensor plates within and sensors outside) connected to a data logger. The data were taken over at least seven days as per existing regulations for historical buildings, and allowed the transmittance of the building materials to be determined.

Data analysis revealed that in relation to the construction techniques with which the walls of the palace were made, the transmittance falls within the limits imposed by law and as such do not consistently modify the variation of internal heat, even in the presence of low outdoor temperatures [10].

As expected, the results showed the lack of thermal bridges. This is because historical building techniques have developed and consistently realized good construction quality with a large thermal mass. The Palazzo dei Capitani is built in travertine on the perimeter walls have a thickness no less than 90 cm. These walls therefore oppose the flow of heat and accumulate a part of it, while maintaining a uniform and substantially constant interior temperature despite very variable external temperatures. Not only does the material have good insulating properties, it can also accumulate heat on one side without releasing it directly to the other, preventing quick external changes in temperature from being reflected directly within the building. Walls with an elevated thermal mass do not disperse the heat within but instead accumulate it during the hottest hours of the day, requiring more time for its dispersion. This contributes to heating and can limit the use of heating systems.

The electrical/mechanical systems in the historical building are particularly varied and complex as a function of the technology used [11]. To assess the efficiency of the systems, it was also important to understand the functional limits of the building in relation to the present uses, which take place over a particularly wide range of time [12]. Point-like analysis of energy consumption was made and analyzed using data provided by the public administration, but it is very difficult to obtain specific data.

One of the problems regards the hours in which the heating systems are working, which, according to law, is considered equal to ten hours per day. This is a valid assumption for residential and service buildings, but risks overestimating the energy consumption of buildings with particular functions, especially if open by appointment or with specific hours.

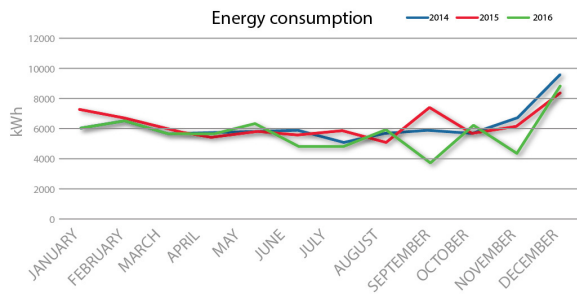


Fig. 6 Energy consumption (2014-2016)

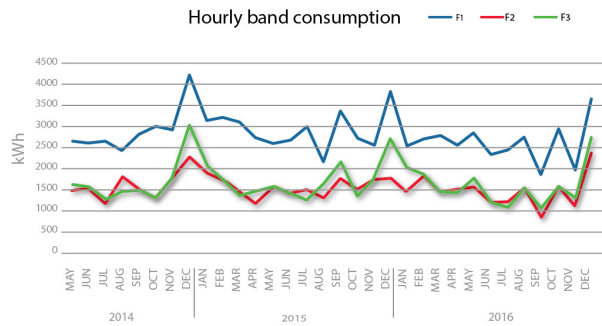


Fig. 7 Hourly band consumption (2014-2016)

	Hour																							
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
M	F3	F3	F3	F3	F3	F3	F3	F2	F1	F1	F1	F1	F1	F1	F1	F1	F1	F1	F1	F2	F2	F2	F2	F3
T	F3	F3	F3	F3	F3	F3	F3	F2	F1	F1	F1	F1	F1	F1	F1	F1	F1	F1	F1	F2	F2	F2	F2	F3
W	F3	F3	F3	F3	F3	F3	F3	F2	F1	F1	F1	F1	F1	F1	F1	F1	F1	F1	F1	F2	F2	F2	F2	F3
T	F3	F3	F3	F3	F3	F3	F3	F2	F1	F1	F1	F1	F1	F1	F1	F1	F1	F1	F1	F2	F2	F2	F2	F3
F	F3	F3	F3	F3	F3	F3	F3	F2	F1	F1	F1	F1	F1	F1	F1	F1	F1	F1	F1	F2	F2	F2	F2	F3
S	F3	F3	F3	F3	F3	F3	F3	F2	F1	F1	F1	F1	F1	F1	F1	F1	F1	F1	F1	F2	F2	F2	F2	F3
S	F3	F3	F3	F3	F3	F3	F3	F2	F1	F1	F1	F1	F1	F1	F1	F1	F1	F1	F1	F2	F2	F2	F2	F3

Fig.8 Hourly band by national regulations

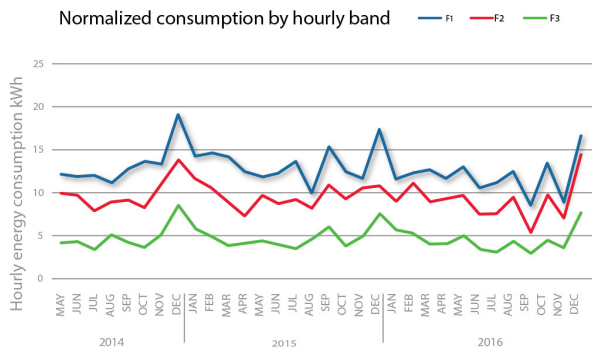


Fig. 9 Normalized consumption by hourly band (2014-2016)

Thermal systems used to heat rooms and prepare hot water for sanitary use are inevitably subject to degradation over time due to the combined action of mechanical and thermal stress and wear, which can reduce the level of safety and efficiency. So that the thermal system functions correctly over time in terms of safety, energy efficiency, and environmental

protection, it should be subject to periodic checks and careful programmed maintenance (Fig. 6).

Over the three years considered (2014–2016), substantial coherence was seen in the consumption trends, which overlap in most of the period. Useful information about the time profile of consumption can be made by analyzing the consumption trends per hourly band (Fig. 7).

To clarify, Italy has national regulations regarding energy consumption from the grid. The day is divided into three hourly periods that are used to define the cost of energy.

From Figure 9, therefore, it is correct to say that most consumption occurs during the day, with an average of 2852 kWh/month consumed during phase F1. To better understand the consumption over time, it is useful to observe the consumption trends normalized by the number of hours falling in the specific phase as a function of time.

The figure 9 shows that the profiles maintain the trends shown in Figure 7, but the scale highlights the fact that the amount of consumption is not uniformly distributed over the three phases and there are distinct differences between F2 and F3.

Briefly, the load profile is very unbalanced during daylight hours, which, in the spring/summer, cover both phase F1 and a large part of F2.

Keeping in mind the available surface area, a photovoltaic system with a power of 25 kWp could be installed directly on the slope of the roof [13]. As is shown in the last line in Table 1, the production would almost completely cover phase F1 (91%). Given the particular load profile, it manages to cover an average of 75% of the needs in phase F1, reaching 100% during the summer months. The lowest coverage obviously occurs in December, when the lowest production during the year occurs during the greatest consumption.

	Production			Consumption			Coverage		
Month	F1	F2	F3	F1	F2	F3	F1	F2	F3
January	1187	28	0	3156	1924	2109	38%	1%	0%
February	1588	82	0	3220	1735	1771	49%	5%	0%
March	2513	155	53	3133	1469	1392	80%	11%	4%
April	2928	195	131	2751	1192	1511	100%	16%	9%
May	3341	227	219	2620	1595	1588	100%	14%	14%
June	3419	229	244	2712	1427	1411	100%	16%	17%
July	3689	239	218	3016	1513	1267	100%	16%	17%
August	3375	213	151	2169	1344	1653	100%	16%	9%
September	2640	162	73	3386	1799	2175	78%	9%	3%
October	2016	111	12	2740	1522	1377	74%	7%	1%
November	1301	40	0	2585	1745	1812	50%	2%	0%
December	1108	17	0	3854	1789	2734	29%	1%	0%
Percentage	91%	5%	3%	47%	25%	28%	75%	10%	6%

Table 1: Production and consumption balance

With respect to the overall electricity needs, the system covers about 38% of the consumption needs, with a percentage of self-consumption equal to 89%. These numbers are decidedly encouraging with respect to the sustainability of this type of approach since the night-time consumption (phases F2 and F3) is not due to work activities and can therefore certainly be minimized. For example, improving the efficiency of the exterior lighting (substituting sodium-vapour lamps with LEDs) would lead to a strong reduction in night-time consumption. Added to this, the loads can be moved both through procedural (turning on backup computers) and technological expedients (both thermal and electrical energy storage systems).

Finally, it should be noted that phase F3 also covers daytime hours on Sunday. Therefore, a further 15% could reasonably be moved to the daytime phase and therefore covered with the photovoltaic system.

The theme of photovoltaic technology within historical town centres relies on very few applicative examples due to the difficulty of combining the architectural good with the insertion of a technology that tends to alter its form and appearance [14].

V. CONCLUSIONS

In the cultural heritage is possible to reduce consumption, increase internal comfort and improve energy efficiency, respecting characteristics and historical value, through a multi-interdisciplinary approach, both theoretical and technological, guaranteeing quality interventions aimed at specific cases.

The energy performance of historic buildings and the possibilities of their improvement depend strictly on the characteristics of each building. The various historical technical solutions, and above all conservation requirements and historical and architectural restrictions, do not allow to identify similar strategies to those used for new buildings.

However, it is necessary to analyze the impact that the features of local historical architecture have on energy performance, to delineate guidelines and above all to facilitate the dissemination to a new approach to the restoration project of the historic building for a more suitable energy efficiency.

The integrated system of information and data necessary to understand the themes tackled should include the range of technical elements and the interaction of buildings with the surrounding context.

Referring to significant cases or representative buildings, it is possible to propose criteria and models that identify admissible energy improvement solutions.

This paper deals with energy saving in a typical historical building of Ascoli Piceno, it offers the opportunity to define a

methodology applied to a meaningful historical context. Energy performance, formalization of a correct framework of requirements and the effective possibility of savings have evaluated from the administration within its development strategies.

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